

ROLL NO : MS NAME :

PHY201 EndSem PART A Date : November 29, 2024 Inst: Abhishek Chaudhuri

- Time : 45 minutes
- Max Marks : $5 \times 3 = 15$
- Attempt all questions.

Please give your answers in the space provided.

1. A plane harmonic infrared wave travelling through a transparent medium is given by

$$\mathbf{E}(y, t) = E_0 \sin 2\pi \left(\frac{y}{5 \times 10^{-7}} - 3 \times 10^{14} t \right) \hat{\mathbf{x}}$$

in SI units. What is the refractive index of the medium?

[3]

2. Consider a forced oscillator with periodic forcing $F = F_0 \cos \omega t$. Sketch the amplitude $A(\omega)$ of oscillation against driving frequency ω when (a) there is damping and (b) when there is no damping. Mark the maxima and appropriate limits on the two axes. Also, plot the variation of phase angle δ with driving frequency when there is no damping.

3. The visibility of an interference pattern is defined as $V = \frac{(I_{max}-I_{min})}{(I_{max}+I_{min})}$, where I_{max} and I_{min} are the maximum and minimum intensities, respectively. If the ratio of intensities of two interfering coherent beams is α , what is the visibility of the pattern in terms of α ? [3]
4. Write down the electric field for a monochromatic plane wave of amplitude E_0 and frequency ω , which is travelling in the direction from origin to the point $(1, 1, 0)$, with polarisation parallel to the yz plane. [3]
5. Two parallel narrow slits in an opaque screen are separated by 0.100 mm. Plane waves of wavelength 600 nm illuminate them. A cosine-squared fringe pattern wherein consecutive maxima are 3.00 mm apart appears on a viewing screen. How far from the aperture screen is the viewing screen?

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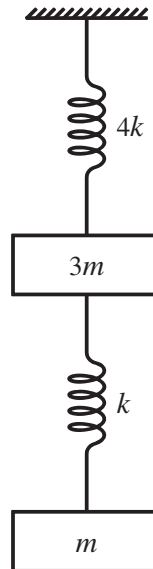
PHY201 EndSem PART B Date : November 29, 2024 Inst: Abhishek Chaudhuri

- Time : 2 hr 15 minutes
- Max Marks : 35

1. Consider the two oppositely travelling electric-field waves, [1+2+3]

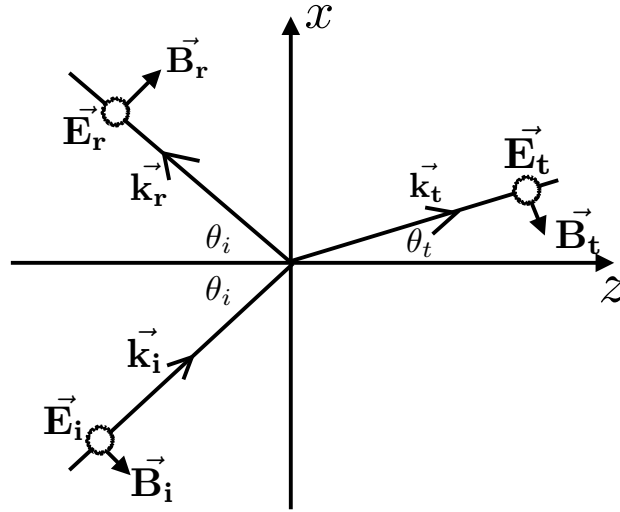
$$\mathbf{E}_1 = \hat{\mathbf{x}}E_0 \cos(kz - \omega t) \quad \text{and} \quad \mathbf{E}_2 = \hat{\mathbf{x}}E_0 \cos(kz + \omega t).$$

- (a) Show that the sum of these two waves is the wave, $2\hat{\mathbf{x}}E_0 \cos(kz) \cos(\omega t)$. What is the nature of the wave?
- (b) Find the magnetic field associated with this electric wave by finding the $\mathbf{B}$ fields associated with each of the above travelling $\mathbf{E}$ fields and then adding them.
- (c) Find the magnetic field by instead using Maxwell's equations to find the $\mathbf{B}$ field associated with the electric wave, $2\hat{\mathbf{x}}E_0 \cos(kz) \cos(\omega t)$
2. A wave of frequency 20 sec^{-1} has a velocity of 80 m/sec . (a) How far apart are two points whose displacements are 30° apart in phase? (b) At a given point, what is the phase difference between two displacements at times separated by 0.01 sec ? [3+2]
3. The figure shows two masses of mass $3m$ and m hanging from springs of spring constants $4k$ and k , respectively. (a) *Setting this up as an eigenvalue problem*, find the normal oscillation frequencies (b) Describe the normal modes. [4+2]



4. A long string of mass 1.0 g cm^{-1} is joined to a long string of mass 4.0 g cm^{-1} and the combination is held under constant tension. A transverse sinusoidal wave of amplitude 3.0 cm and wavelength 25 cm is launched along the lighter string. (a) Calculate the wavelength and amplitude of the wave when it is travelling along the heavier string. (b) Calculate the fraction of wave power reflected at the boundary of the two strings. [3+3]

5. An incident wave whose polarization is *perpendicular* to the plane of incidence (say $x - z$ plane) falls on the boundary between two linear media at an angle θ_i . Then, the reflected and the transmitted waves are also polarised in the same $x - z$ plane.



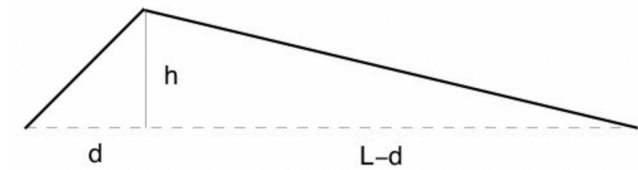
- (a) Obtain the reflected and transmitted amplitudes in terms of incident amplitude. [4]
 (b) Find the *reflection coefficient*, R , and the *transmission coefficient*, T . [2]
6. Consider a string of length L fixed at both ends. The complete expression for the transverse displacement of the string, as a function of time and position along the string's length, is

$$y(x, t) = \sum_{n=1}^{\infty} [A_n \cos(\omega_n t) + B_n \sin(\omega_n t)] \sin\left(\frac{n\pi x}{L}\right)$$

where ω_n is the natural frequency of the n -th mode, and the coefficients A_n and B_n are determined from the initial conditions. Suppose the string is plucked, as shown in the figure. It is pulled to a transverse distance h at a distance of d from one end and then released from rest. The initial shape of the string is triangular, comprised of two straight line segments:

$$y(x, 0) = \begin{cases} h \frac{x}{d} & \text{for } 0 \leq x \leq d, \\ h \frac{L-x}{L-d} & \text{for } d \leq x \leq L \end{cases}$$

- (a) Write down the complete Fourier Series expression for the motion of the plucked string. (b) Sketch the



- motion for a few time points. (c) What time signal would a guitar pickup record when a string is plucked? In other words, what is the time behaviour of a specific location on the string? [4+1+1]